

**DentiSys: A Web-based Academic Grading and Retention Monitoring System for
the College of Dental Medicine at Bicol University**

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1 INTRODUCTION

1.1 Background of the Study

The continuous growth of information technology has changed the way educational institutions manage academic records, monitor student performance and attendance efficiently. Existing systems that automate grading and enable easier management and accessibility of student information have shown to enhance accuracy, improve efficiency and decision-making (Bato & Pomperada, 2025). Compared to traditional paper-based and manual entry methods which often cause delays, are less efficient, and not easily accessible, these technologies are able to provide real-time insights on student performance to make timely academic interventions (Akash, Kingsley & Murali, 2024). As a result, institutions are better equipped to streamline academic processes, improve transparency and accessibility to student information.

Academic policies, specifically retention policies, are a critical factor in maintaining educational standards in the Philippines. Higher education institutions are required to enforce strict admission and retention policies for professional programs such as Dental Medicine (Commission on Higher Education, 2019). While these policies prove to be effective, there are studies suggesting that students often have a hard time complying due to various academic and personal challenges (Ablian et. al, 2023) and may cause a negative impact on the students which potentially leads to disqualification. Therefore, there is a need for effective systems that can monitor academic performance, at the same time provide timely interventions for those at risk, and support retention requirements.

Attendance tracking is another critical factor in a student's academic performance that requires system improvement. Traditional methods that rely on paper-based monitoring are often prone to inefficiencies, proxies, and data loss. As a result, technology assisted attendance systems have been developed to counter these problems (Zainab, 2018). While various digitized attendance systems are more efficient compared to traditional attendance systems, there are still apparent disadvantages to each of these methods. QR and RFID-based systems may introduce delays and be more susceptible to proxy attendances. While fingerprint-scanners that have been shown to reduce proxy attendances are not cost-effective and quite difficult to implement (Orijakor et al., 2025). On the other hand, while facial recognition is more often used in professional settings, it can also be implemented for educational purposes. It has also shown to be practical and cost-effective, requiring only minimal hardware to be applied (Vadwala, 2024). Through facial biometric authentication, attendance records become more reliable as physiological attributes are difficult to replicate (Mir et al, 2018). Furthermore, face recognition-based monitoring systems have proven to be efficient in attendance recording, while minimizing manual intervention (Ramesh, 2020).

Globally, many studies have developed systems that aim to improve academic processes in different educational institutions. For instance, an e-class record system has been shown to enhance efficiency in computing grades, monitoring student performance, and reducing manual work (Guirit, 2023). In the Philippines, automated grading systems with performance analytics are being normalized nowadays to improve education and streamline tasks. These systems were developed to tackle the challenges and inefficiencies produced by traditional grading systems (Bato & Pomperada, 2025). A customized grade evaluation and student retention system for

Eastern Samar State University demonstrated reliability and efficiency in evaluating grades and monitoring retention status of students in compliance with the university's retention policies (Cantilang, 2019). Despite these advantages, many existing systems remain limited to grade computing and student performance tracking with minimal integration of retention and attendance monitoring functionalities.

At Bicol University, though the institution offers a variety of programs and implements a standardized grading system, the Dental Medicine program of Bicol University, in particular, shows several gaps in their current system. Their existing grading process relies on Microsoft Excel or Spreadsheets as its main system making it less efficient and prone to errors. Professors also utilize individualized grading sheets which may lead to inconsistencies due to the absence of a centralized grading system. Additionally, the College of Dental Medicine strictly implements their retention policy, however the reliance on manual encoding makes it difficult and time-consuming for professors to monitor and update student status. This not only affects the faculty, but also affects students in tracking and complying with retention requirements.

Furthermore, although attendance is not directly included in their grading criteria, it indirectly affects academic outcomes. Most professors would conduct tests at the end of the lecture, and in such cases, problems arise when a student who was absent on the day of the test would receive transmuted grades that do not reflect their performance. For instance, an absent student who should receive a score of zero may instead receive a transmuted grade equivalent to the lowest passing grade leading to inconsistencies. This emphasizes the need for a reliable integrated system for attendance and academic performance monitoring.

To bridge these gaps, this study proposes “DentiSys: a Web-Based Academic Grading and Retention Monitoring System for the College of Dental Medicine of Bicol University”. This system aims to provide a centralized and efficient system to accommodate BUCDM’s customized grading system and retention policy.

1.2 Objectives of the Study

The study proposes an academic grading and retention monitoring system for the professors of the College of Dental Medicine of Bicol University, which aims to automate and standardize grading methods, retention monitoring and student grade management. Specifically, the study aims to:

1. Design and develop a responsive, secure, and robust system which encompasses:
 - a. A Grade Computation Module that integrates transmutation formulas and weighted grading structures;
 - b. A Retention Monitoring Module that tracks remedial action and predicts retention risk using a Random Forest Algorithm; and
 - c. An automated attendance monitoring system using real-time facial recognition.
2. Include a Report Generation Module that produces:
 - a. Comprehensive grading sheets;
 - b. Attendance sheets; and
 - c. Summaries of students that require remedial action.

3. Evaluate the system to be developed using ISO/IEC 25010 software product qualities in terms of functional suitability, portability, reliability, maintainability, compatibility, performance efficiency, security, and usability.

1.3 Significance of the Study

In light of BUCDM's need for operational efficiency through automation, this study aims to create a system specifically designed to accommodate the college's unique grading system and retention policy. Furthermore, this system will be beneficial to the following:

Bicol University College of Dental Medicine. By automating grading, retention policy implementation, and attendance tracking, the system ensures institutional policies are applied uniformly and accurately across the college.

Faculty. To assist faculty members' day-to-day operations. Features such as automated grade calculation, retention status flagging, and attendance tracking aim to expedite and reduce the repetitive manual labor required in student grading and retention monitoring, which allows faculty members to allot more time into teaching, mentoring, and improving their quality of education.

Students. The study provides students with a means to log class attendance through facial recognition technology, which automates the tedious task of listing names, passing attendance sheets, or doing a standard roll-call. Moreover, by utilizing this technology, the system significantly reduces the possibility of proxy attendance or human error.

Future Researchers. Shall serve as a reference for future studies in the realm of academic information systems, specifically those integrating grading automation, retention monitoring, and biometric attendance tracking.

1.4 Scope and Delimitations

This study focuses on the development of DentiSys, a web-based academic grading and retention monitoring system with automated attendance monitoring through facial recognition for the College of Dental Medicine of Bicol University.

The system covers the design and development of a Grade Computation Module that allows professors to input, store, and compute student scores and grades for both lecture and laboratory components. The module will support specific transmutation formulas and allow professors to customize grading weights according to their respective grading structures.

The system includes a Retention Monitoring Module that tracks remedial action and identifies students who may be at risk of failing to meet retention requirements. This module will use a Random Forest machine learning technique to predict student retention risk based on encoded academic data and relevant academic indicators aligned with the College of Dental Medicine's retention criteria.

The system provides report generation features that support academic monitoring and documentation. These include attendance sheets, student summaries, and student grade-related reports that may help professors review student performance and retention status.

The system features an automated attendance monitoring component using real-time facial recognition. The attendance module will use an external camera to automatically capture student facial data during classroom attendance checking. The captured facial data will then be compared with stored student facial data using an existing pre-trained facial recognition model for attendance verification.

The developed system will be evaluated using selected ISO/IEC 25010 software product quality characteristics, specifically functional suitability, portability, reliability, maintainability, compatibility, performance efficiency, security, and usability, to determine whether it meets its intended functions and is acceptable for use by its target users.

However, the system is limited to implementation within the College of Dental Medicine of Bicol University and does not include university-wide deployment, cross-campus integration, integration with existing learning management systems or other institutional systems, or migration of historical academic records. The grading and retention monitoring components are limited to professional courses under the College of Dental Medicine's retention policy. The facial recognition component is limited to classroom attendance monitoring only and is not intended for surveillance, general identity authentication, or monitoring outside instructional attendance. Attendance records generated through facial recognition may still be subject to verification and correction when necessary. Additionally, the retention risk monitoring component is limited to risk identification based on encoded academic data, thresholds, and weighted academic indicators, and will not replace official academic evaluation, faculty judgment, or administrative decision-making.

2 THEORETICAL FRAMEWORK

This section contains the foundational literature used to conceptualize, support, and bring the proposed system into fruition.

2.1 Review of Related Literature

2.1.1 Digitized Grading Systems

Academic grading systems have been part of our educational institutions, and one of the major concerns still remains, the management and processing of it. Grading systems in higher education face several challenges, including time inefficiency, subjective bias, and scalability (Deepshikha, 2025), which highlight the need for more effective solutions. In a similar manner, Tan et al. (2025) examined traditional manual grading in relation to automated systems and found that automated systems improve consistency and scalability while also reducing educators' workload. These observations point to the need for a transition from manual grading practices toward more efficient and reliable automated alternatives.

Early digitized tools such as e-class records were introduced to support educators in managing grades. These systems improve the reliability and efficiency of grading student performance by automating grades computations (Gurit, 2023). Similarly, the Department of Education implemented standardized e-class records that have been shown to increase speed and accuracy of grade computation while also reducing common issues associated with manual encoding (Lee, 2020). However, while this tool shows significant improvements from traditional methods, most of these systems are excel-based systems leading to data silos. This fragmentation makes it

hard for educators and administrators to obtain a comprehensive view of student performance.

As educational institutions continuously adapt and improve, centralized web-based student information systems were developed as these systems not only facilitate academic management but also support institutional decision-making (Gürkut & Nat, 2017). By utilizing a web platform, institutions can centralize different academic components into a unified platform. Ramadhani et al. (2025) and Doctor (2017) state that such systems have been shown to improve accessibility, allowing educators and the administrators to input, manage, and view data. Despite these advancements, many systems still lack the integration of other academic processes such as retention policies and risk monitoring.

To bridge this gap, systems must also comply with grading policies, particularly in the computation and standardization of grades. Grade transmutation systems convert raw scores into standardized grades based on grading policies established by the Department of Education (DepEd), which are adapted by higher education institutions and are tailored to fit their own grading schemes and ensure consistency across varying programs (Forca et al., 2019). Moreover, retention policies are commonly implemented in higher education institutions and are particularly strict in professional programs, such as the Dental Medicine program, to maintain academic standards (Commission on Higher Education, 2018). At Bicol University, the College of Dental Medicine follows a rigorous retention system that students must go through to take the board exam and graduate. Additionally, performance monitoring and analytics are included to analyze student performance and identify failing or at-risk students, thereby supporting data-driven decisions and providing appropriate academic interventions.

To support this, Viberg et al. (2018) found that learning analytics can help improve learning practices to enhance student learning and performance in higher education. Similarly, Robinson & Song (2019) claimed that student performance monitoring systems help track student progress, at the same time enhancing student learning outcomes through timely interventions. Finata & Andrawina (2024) emphasized that performance monitoring in higher education is essential in identifying student's difficulties in the early stages. By tracking outcomes in homeworks, quizzes, recitations, and major exams, it reduces the likelihood of students to retake the course, hence increasing student success rates. Incorporating performance monitoring and analytics tools into academic grading systems helps improve academic planning and data-driven decision-making, all while supporting student development through consistent feedback (Dayanghirang & Hernandez, 2022).

Overall, the literature shows a shift from traditional and semi-digital systems towards modern academic grading systems, and eventually in centralized grading systems that proves there are still inefficiencies in existing systems. The growing importance of integrating grading systems, retention policies, and performance monitoring analytics into a unified platform vaguely emphasizes the gap in current systems. In line with this, the literature collectively highlights an advanced and integrated system that goes beyond automation. Such a system should support real-time academic monitoring, data-driven decision-making, and timely academic interventions, which aligns with DentiSys' objectives that aims to provide a centralized system to accommodate BUCDM's grading and retention policies.

2.1.2 Retention Policy Implementations

Student retention is defined as the ability of a university to enforce policies that ensure that a student remains in the institution and completes all levels of a degree or program based on student academic performance. (Cantilang, 2019).

The retention policy for BUCDM states a student who gets a grade of 2.5 and below is required to take a remedial exam to be retained for the subject. If the student gets 50% on the remedial exam, they are retained in the course. However if they fail to get 50% or above on the remedial exam, they will have to take another remedial exam. If the student fails the second remedial exam, they must re-enrol in a cost recovery program wherein they must get a grade over 2.4. If the student fails the cost recovery program, they are no longer retained in the course.

There is demand in higher education settings for technological solutions to not only enhance the learning environment but also support students. In particular, student retention has always been an area of study that institutions focus on because of the complex factors that influence the issue (Villano et al., 2018). Early Alert Systems (EAS) are essential technologies that are used in certain universities and institutions to help maximize student retention and improve learning outcomes (Harrison et. al 2021). The early identification of at-risk students significantly improves the ability of institutions to administer timely academic intervention and support for students (Matz et al., 2023). Retention monitoring systems enable data-based academic intervention by examining academic indicators that identify which students require academic support (Cele, 2021). In fact, the availability of digitalized student information has allowed many opportunities for academic institutions in identifying at-risk students and analyzing the factors that lead to retention failure. (Vaarma & Li, 2024).

Machine learning models are widely used in predicting student retention and identifying at-risk students using academic and institutional data. (Matz et al., 2023). To support the implementation of the student retention risk classification functionality of the system, the module incorporates a machine-learning based module that utilizes the Random Forest algorithm. The Random Forest model is a type of decision tree model that averages the predictions of individual decision trees to make predictions. (Schonlau et. al, 2020). The algorithm has been shown to outperform other machine learning models in identifying at-risk students. (Tang et al. 2025). In contrast to models such as linear regression which assumes linear outcomes, the random forest algorithm is better suited for medium to larger datasets because it can make nonlinear assumptions about the data, which makes it more reliable at predicting outcomes (Schonlau et. al, 2020).

The features that the algorithm considers are based on the lecture and laboratory grading components of BUCDM which consists of term exams, quizzes, recitations, outputs, practical exams, lab exercises, quizzes, and lab performance. These features are used to predict whether a student is low, moderate, or high risk, which are classifiers that are based on the retention requirements of the college. The use of similar academic indicators have been used to provide a reliable indication of student performance and retention risk in higher education environments. (Matz et al., 2023). Furthermore, the use of multiple academic indicators increases the effectiveness, accuracy, and reliability of the evaluation of at-risk students. (Matz et al., 2023).

2.1.3 Facial Recognition-Based Attendance Systems

Attendance systems in academic institutions encourage punctuality, enhance learning outcomes, and improve education levels (Ali et al., 2019). Although BUCDM

faculty generally do not attribute standalone grades towards attendance, it is a prominent factor in academic success; current transmutation policies maintain that students receive a 0% mark for quizzes missed due to unexcused absences. Evidently, this policy raises a need for a highly accurate, integrated attendance system that consolidates quiz grading to student attendance records.

Generally, there are two types of attendance systems: Manual Attendance Systems (MAS) and Automated Attendance Systems (AAS) (Sawhney et al., 2019). MAS encompasses pen-and-paper registers, roll-calls, and other physical record-keeping methods, while AAS includes biometric, Radio Frequency Identification (RFID), and online attendance systems, among others (Qureshi, 2025). Most academic institutions, including BUCDM, use the former method. However, despite its utility, MAS has proven to be time-consuming, tedious, and inaccurate due to its susceptibility to proxy attendance (Smitha et al., 2020). These seemingly small issues augment in the realm of academe—an industry that puts accuracy and accountability at its forefront (Xhuxhi, 2026). Fortunately, employing AAS nullifies these problems by providing a more efficient, reliable, and accurate alternative to attendance management (Agrawal & Tripathi, 2024).

Several AAS have been proposed to reduce the use of MAS. However, some come with additional burdens to the users. For instance, smart cards, tokens, and cryptographic keys, which employ the use of passwords and personal identity numbers, are accompanied by the heavy cognitive load of having to remember them; in addition, possessed-object-based attendance systems carry the risk of being lost, stolen, or duplicated by unauthorized entities (Bahjat, 2026; Naz et al., 2017). Furthermore,

biometric—specifically, facial recognition—based attendance systems nullify these challenges, for physiology cannot be forgotten or misplaced.

Gururaj et al. (2024) define facial recognition as a type of technology that allows systems to identify and verify the identity of individuals through facial characteristics. However, the collection of such data raises significant privacy and security concerns, as biometric data is inherently sensitive. The National Privacy Commission (NPC) Advisory Opinion No. 2017-063 asserts that a photograph is considered a unique identifier that can map directly to an individual. Unlike passwords that can easily be changed, compromised biometric data carries the burden of being nearly immutable. Therefore, NPC Advisory Opinion No. 2019-049 stipulates that using facial recognition technology must have a lawful basis under the Data Privacy Act of 2012, such as informed consent or legitimate interest, and should adhere to the principles of transparency, legitimate purpose, and proportionality. Furthermore, NPC Advisory Opinion No. 2023-025 emphasizes that security measures such as Role-Based Access Control (RBAC), and encryption are essential for negating the risks associated with facial recognition technology.

To fulfill the aforementioned need for encryption, Sarker et al. (2024) used the Advanced Encryption Standard (AES) algorithm to encrypt images in their open-source facial image quality checker, which they designed for usage in biometric authentication systems. Using AES-128 before storage of the images captured by their system ensured that, in the unfortunate event of a database breach, the sequence of unreadable bytes stored in the database cannot be reversed or reconstructed by an unauthorized entity, thereby enhancing security and preventing spoofing. By adopting these safeguards

alongside RBAC and data minimization, DentiSys may ensure that convenience does not come at the expense of privacy.

Aves et al. (2023) developed an automated academic attendance system leveraging facial recognition technology through OpenCV. The operational workflow of their system during face-to-face classes is initiated by an instructor activating a camera that automatically detects and logs students as they pass within its field of view. Through an ISO/IEC 25010 evaluation system, their study proved to be effective, efficient, and positively received by users. Hence, serving as a robust precedent for biometric integration in local schools and provides an adoptable methodology for implementing facial recognition technology.

Recent literature describes the architecture of traditional computer vision systems as sequential in nature. As demonstrated in the facial recognition attendance system deployed by Gujar et al. (2025), this architecture typically begins with image capture, wherein real-time images are continuously acquired through a webcam or camera module. Then, during the image preprocessing stage, images are normalized using Histogram Equalization and converted into grayscale to improve the consistency of feature extraction under varying environmental conditions. The subsequent stage involves face detection and recognition, wherein the researchers used pre-trained Convolutional Neural Networks (CNN) for both tasks. Their technical stack consisted of React.js, Tailwind CSS, and Redux for front-end development; Flask and Python for back-end development; OpenCV and MTCNN for face recognition; and MySQL for database management and storage. Aligning with these established frameworks, DentiSys will observe the aforementioned foundational framework, integrate Python with OpenCV for real-time capture and processing, implement React.js and Tailwind CSS for

front-end construction, utilize Python and PHP for back-end control, and employ MySQL for its database management system (DBMS).

In terms of facial recognition approaches, Guo et al. (2024) ascertained through a thorough comparative analysis of more than 30 datasets that while 3D based facial recognition offers benefits such as invariance to lighting inconsistencies and cosmetic use, 2D recognition remains more widely adapted for its accessibility, affordability, and ease of implementation. This study supports the feasibility of the proposed system's use of 2D based facial recognition analysis within a resource-constrained academic environment.

Further supporting this approach, Cañete (2021) and Jadhav et al. (2025) successfully implemented AI-powered facial recognition attendance systems with the use of traditional computer vision algorithms, specifically Haar Cascade Classification and Local Binary Pattern Histogram (LBPH). These systems were evaluated through parameters such as accuracy and efficiency; notably, both systems demonstrated high performance, with Cañete reporting 100% accuracy and Jadhav et al. achieving between 95% to 97%. However, a comparative analysis between traditional methods and deep learning facial recognition algorithms by Zeng (2024)—wherein seven (7) varying datasets with more than a million images were used—concluded that deep learning algorithms such as CNNs outperformed traditional algorithms like Local Binary Patterns (LBP) in accuracy by nearly 10%, with its success being largely attributed to its resilience to different environments, lighting variance, and facial occlusion. Nevertheless, traditional algorithms remain highly reliable under controlled conditions. Consequently, given that the proposed system will be deployed in a structured environment with consistent lighting and limited angle variability, the use of traditional computer vision

techniques is deemed more appropriate. This choice ensures lower system requirements and practical feasibility for academic settings without sacrificing necessary performance.

Overall, the literature highlights the shift from MAS to AAS to resolve inefficiencies and attendance fraud in academic systems. Among the myriad of solutions available, 2D facial recognition—specifically, with the use of Haar Cascade and LBPH—proves to be a highly feasible choice for a resource-constrained environment such as BUCDM due to its reliability in controlled conditions. However, the transition to biometric systems emerges data privacy concerns that shall be reconciled through AES-128 encryption, RBAC, and adherence to National Privacy Commission (NPC) guidelines. Ultimately, by aligning with established architectural sequences (acquisition, preprocessing, and detection) and utilizing a modern technical stack of Python, OpenCV, and MySQL, the DentiSys project can provide a reliable, privacy-compliant, and efficient solution for academic attendance monitoring.

2.2 Gap Bridged by the Study

As technology continues to advance, academic institutions have integrated systems in order to improve academic operations. Guirit (2023) found that e-class record systems can decrease manual workload and increase faculty grade computation efficiency. Bato and Pomperada (2025) explained that automated grading systems with performance analytics can support data-driven interventions by allowing educators to track student progress. On the retention side, Cantilang (2019) showed that digital retention monitoring is both practical and reliable. Attendance systems using facial recognition have also been shown to be more accurate and harder to manipulate compared to traditional methods (Mir et al., 2018; Ramesh, 2020). However, these

systems mostly address only one concern at a time and are rarely built around the specific policies of a particular institution.

The College of Dental Medicine of Bicol University still manages grading through individual Microsoft Excel sheets, monitors retention manually, and does not record attendance. These separated processes create inconsistencies and increase the risk of errors. There is no existing system reviewed in this study that attempts to combine all three into a single platform while also conforming to the college's specific grading system and retention policy. This study aims to address that gap through the development of DentiSys. Rather than forcing the college to adapt to a generic system, DentiSys was built around how BUCDM actually operates, from its grading distribution and transmutation formula to its multi-stage retention process and policy on absent students receiving a zero without transmutation.

2.3 Concept of the Study

The conceptual framework of this study was obtained from the input-process-output model, as shown in Figure 1, where two distinct inputs joined into a single unified system. This dual-input structure is grounded in systems theory (Mele, Pels & Polese, 2010), which holds that interrelated but distinct components must work together toward one unified goal, reflecting how the grading system and attendance system, though different in nature and data source, are designed to operate as one cohesive platform. The framework is further supported by data-driven decision-making (Marsh, Pane & Hamilton, 2006) and biometric authentication theory (Abdul-Al et al.,

2024), to ensure that the system is not only functional but also based in established academic thought.

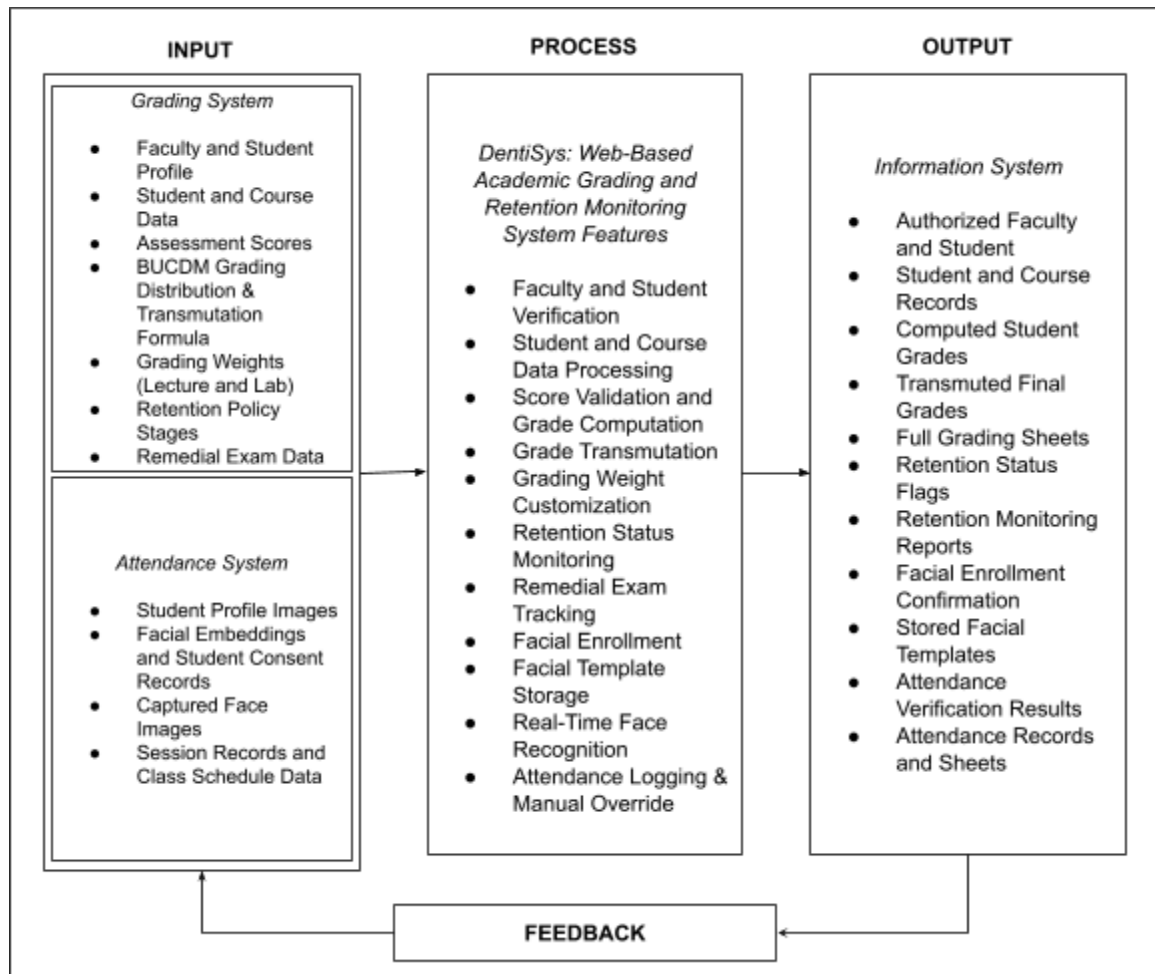


Figure 1. Conceptual Framework

The input phase consisted of two components. The first input covers the grading system, which included faculty and student profile, student and course information, assessment scores, the college's official grading distribution and transmutation formula, grading weights for lecture and laboratory components, retention policy stages, and remedial exam data gathered through interviews with the dean and faculty, as well as a thorough review of BUCDM's existing policies. The second input covers the attendance

system, which included student profile images, facial embeddings and student consent records, captured face images, and session records and class schedule data enrolled prior to system use. This follows the idea of systems theory (Mele, Pels & Polese, 2010), which states that components do not work alone, they are connected and work toward one goal. Instead of treating grading, retention, and attendance as separate concerns, the system brings them together into one organized and cohesive platform.

The process phase contains the events of the methodology that were used for the development of the system. It describes what the system should do for both the faculty and students information. The process included faculty and student verification, student and course data processing, score validation and grade computation, grade transmutation, grading weight customization, retention status monitoring, remedial exam tracking, facial enrollment, facial template storage, real-time face recognition, attendance logging and manual override. This phase is based on the framework of data-driven decision-making (Marsh, Pane & Hamilton, 2006), which holds that systematic data collection and analysis should guide academic decisions to improve student and academic outcomes. By automating grade computation and retention monitoring, the system enables faculty to identify at-risk students early and make timely, evidence-based interventions rather than relying on manual and less reliable processes. Also, biometric authentication theory (Abdul-Al et al., 2024) establishes that physiological characteristics such as facial features are among the most reliable and difficult to manipulate forms of identification, supporting the use of facial recognition as a more accurate and secure alternative to traditional attendance monitoring methods.

Finally, the outputs include authorized faculty and student records, student and course records, computed student grades, transmuted final grades, full grading sheets,

retention status flags, retention monitoring reports, facial enrollment confirmation, stored facial templates, attendance verification results, and attendance records and sheets, all accessible through a centralized web-based interface. After the output, the feedback regarding the completed system was necessary for the improvements of the study and system.

2.4 Definition of Terms

For a better comprehension of the concept presented in this study and to further understand the used key terms, the following important terms are defined:

Academic Grading System. This is a system used in education institutions to manage, record, and compute student academic performance.

Attendance Monitoring System. This is a system designed to track and manage student attendance.

Centralized System. This is a system architecture in which all data and processes are managed in a unified platform, integrating grading, attendance, and retention functionalities within a single system.

DentiSys. The proposed grading and retention monitoring system for the College of Dental Medicine of Bicol University.

Facial Recognition. The biometric technology used to detect and recognize human faces for the attendance monitoring module.

Faculty. These refer to the college's academic staff—encompassing instructors, teachers, and professors.

ISO/IEC 25010. This refers to a software quality standard assurance that defines a set of requirements and characteristics for software quality evaluation, including functionality, reliability, usability, and security.

Local Binary Patterns Histograms (LBPH). A traditional computer vision algorithm used for facial recognition that is primarily provided by the OpenCV library. It works by comparing pixel neighbors to identify patterns, and using histograms to represent facial features.

Professional course. Within the BUCDM curriculum, these refer to major, program-focused, practical training classes designed to prepare students for the practice of Dental Medicine. These courses require a minimum grade of 2.4. Failure to meet this requirement triggers a series of interventions that serve as a determining factor for student retention in the program.

Retention Monitoring. This is a module that tracks student retention status and identifies students who may be at risk of failing to meet retention requirements.

Retention Policy. This is an institutional guideline applied in professional programs that determines a student's eligibility to continue in a degree program based on academic performance.

System Module. This is a functional component of a system that performs specific functions such as grading, risk flagging, and attendance monitoring module.

Transmutation. This is the process of converting raw scores into standardized grades based on institutional grading policies.

3 MATERIALS AND METHODS

This section presents the materials and resources utilized in the development of the proposed system, as well as the approach taken in software development to achieve the objectives of this study.

3.1 Materials

This section discusses the software and hardware materials, along with the version requirements used in the documentation of the study and the development of the proposed system.

3.1.1 Software

The software tools and programs utilized in the development, implementation, and deployment of the proposed system are listed in Table 1 with their corresponding software versions.

Visual Studio Code was used as the primary code editor due to its familiarity and support for the various programming languages used in the project.

MariaDB was selected as the database management system due to its open-source nature, improved performance, and compatibility with PHP, making it a preferred choice for developers.

PHP was chosen as the primary back-end programming language due to its simplicity, ability to integrate databases, and suitability with web-based systems.

To support specific system functionalities and data processing, Python was utilized as a development stack due to its versatility and broad library support.

Table 1. Software Versions

Components	Software	Description
IDE	Visual Studio Code 1.117.0	Utilized for developing and editing the source code.
Database	MariaDB 10.4.32	Stores and manages system's data and database transactions.
Server-Side Language	PHP 8.2.12	Processes system functions, requests, and communicates with the database.
Front-end Stack	HTML5, JavaScript, Tailwind CSS 4.2.2, React.js 19.2.	Responsible for creating system interface and improving user interaction within the system
UI/UX Design Tool	Balsamiq	Designs screen layouts and system prototype before development.
Back-end Programming Language	Python 3.14.3	Utilized for additional system functions during development
System Version Control	Git 2.51.1	Tracks changes in the source code and supports collaboration.
Repository Platform	GitHub	Serves as the online repository for storing and managing project files.
Hosting Platform	Docker 4.70.0	Provides a platform for deployment and system testing.

Before writing the code for the front-end development, Balsamiq was utilized to design the system's user interface wireframes and prototypes, providing visualization and flow of interaction.

HTML5 and JavaScript were used to provide structure and interactivity to the system, while Tailwind CSS, a front-end framework, was used to make the design process of the interface faster, consistent, and more efficient. Additionally, React.js was used as a primary JavaScript library in developing a dynamic user interface.

Git was used as the primary version control system to track changes in the source code and accommodate changes of each member, while GitHub served as the cloud-based repository to manage the project, facilitate collaboration, and track version changes. Furthermore, Docker was used as a hosting platform for the proposed system to ensure consistent deployment across different environments and devices.

These software tools and programs were selected to ensure optimal efficiency and collaboration during the development process.

Table 2. Algorithm and Corresponding Module

Algorithm	Corresponding Module
Local Binary Patterns Histograms (LBPH)	Attendance Monitoring
Haar Cascade Classification	Attendance Monitoring
Random Forest Model	Retention Monitoring

The implemented algorithms for the retention and attendance monitoring module are summarized in Table 2. Local Binary Patterns Histograms (LBPH) and Haar Cascade Classification, traditional computer-vision based face recognition algorithms are

utilized for the Attendance Monitoring module to help automate attendance tracking through facial recognition. This technique enhances efficiency and reliability in managing attendance. While Random Forest is utilized for the retention monitoring module to track retention status and predict students at-risk. These algorithms were selected to ensure reliability and accuracy of the system modules.

3.1.2 Hardware

This subsection discusses the hardware components that are required to launch, run, and maintain the system. The system uses a client-server architecture to facilitate internal operations such as database access, retention risk analysis, and client-server requests. The processes are handled inside the private network of the university. The server is crucial in this architecture because that is where the data is stored centrally. It requires higher specification than the client platforms. In contrast, desktop and mobile platforms require lower specifications, and often only require stable network access.

The specifications of the local server are kept at a moderate to high level to ensure that operations run seamlessly. The server has a RAM of at least 8GB to make sure that multiple concurrent users can access the system reliably, database operations run smoothly, and multitasking is done efficiently. The server also uses SSD storage to improve data reading and writing times and to reduce latency in data retrieval and report generation process. Furthermore, a stable LAN connection with a speed of at least 25 Mbps is also essential to ensure that the data can be accessed in real-time effectively.

The desktop and mobile platform specifications are kept to a safe minimum since most of the processing is done on the internal server. For desktops and laptops, the minimum requirements are an Intel i3 CPU or a Ryzen 3 CPU, along with a minimum of

4GB of RAM. This ensures that the client device can reliably access the system interface, the dashboard, the student grading module, the retention prediction module, and the rest of the system's functionality. For mobile devices, the operating system required is either Android 8 and above or iOS 12 or above. The RAM for mobile devices is 2GB, which also ensures access to the basic functionalities of the system. For both desktop and mobile devices, a storage capacity of at least 1GB or more will suffice since most of the storage usage comes in the form of cache and offline storage. The client devices also require a stable internet connection to ensure that access to the system remains uninterrupted.

Table 3. Hardware Specifications

Components	Processor/ OS	RAM	Storage	Internet Speed	Screen Resolution
Physical Server	Quad-Core	8GB	256 GB	25 Mbps	—
Personal Computers (Desktop and Laptops)	Intel Core i3 or Ryzen 3	4GB	1GB	25 Mbps	1366x768
Mobile Phones	Android 8 / iOS 12	2GB	1GB	25 Mbps	at least 5 inches

3.2 Software Methodology

This study adopted the Rapid Application Development (RAD) model, as illustrated in Figure 2, as its primary Software Development Life Cycle (SDLC) model for the development of the proposed system. As described by Sommerville (2016), RAD is an incremental software development approach that prioritizes rapid prototyping and continuous user feedback over lengthy planning and documentation, allowing

development teams to build and refine system components quickly through active stakeholder involvement.

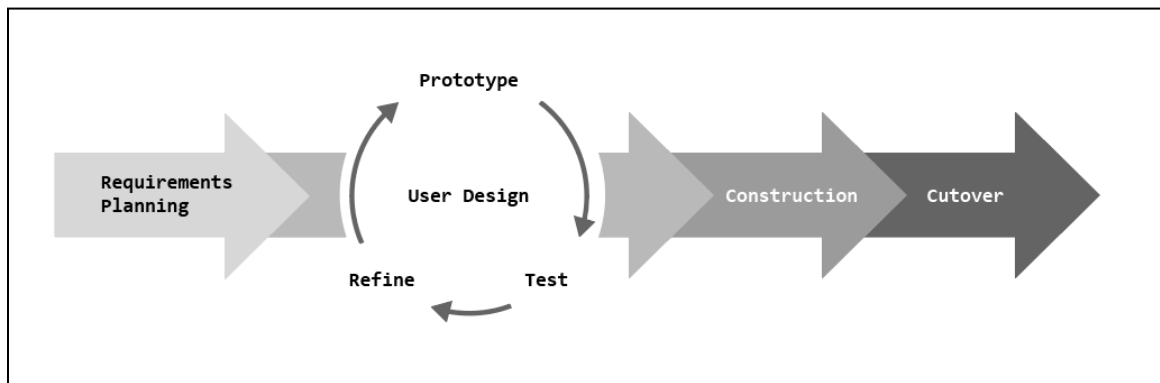


Figure 2. Rapid Application Development (RAD)

RAD was selected because the development of the system required continuous collaboration and iterative refinement of system components throughout the development cycle. While the core policies of the Bicol University College of Dental Medicine, including their grading distribution, transmutation formula, and multi-stage retention policy, served as the foundation of the system's requirements, the complexity of translating these institutional policies into functional system modules necessitated an approach that allowed for rapid prototyping and regular validation with the faculty at every stage of development. RAD's prototype-based structure enabled the development team to build, test, and refine each module incrementally, ensuring that the system accurately reflected the actual needs of the college.

While RAD is sometimes critiqued for being difficult to manage when user involvement is inconsistent throughout the development cycle, this will not pose a significant risk for this study. This limitation will be mitigated by establishing scheduled consultation and prototype review sessions with the BUCDM faculty and administration at key stages of the project, ensuring that feedback is gathered systematically and that

each development phase remains focused, purposeful, and aligned with the college's institutional needs.

3.2.1 Requirements Planning

The Requirements Planning Phase is integral to the correctness, completeness, and success of a system. It is foundational; this phase entails stakeholder collaboration, such that the project's necessary functionalities may be defined. The system requirements of DentiSys were established through a semi-structured interview and questionnaire with the Office of the Dean at BUCDM.

The interview with the Dean of BUCDM assessed preliminary information necessary for building the initial scope of DentiSys. The session detailed the college's grading system, retention policy, compliance documents, current processes, and its challenges. Information gathered from the interview was further supplemented by assistance from the Associate Dean in viewing supporting documents such as BUCDM's retention policy flowchart (see Appendix C) and a quick perusal of a grading sheet template. Further questions and requests had been made through asynchronous digital communication to accommodate their administrative workload. Specifically, a mixed questionnaire was utilized to ask for further clarifications about grading practices, request formally documented policy, acquire feedback regarding proposed features, and gauge preferences for an attendance system.

The transcripts created from these two (2) sessions then served as groundwork for the establishment of the project objectives, scope, requirements, and design. In

addition, it enabled the researchers to determine tools and technologies needed to actualize the identified project objectives.

3.2.2 User Design

The design phase of DentiSys focused on converting the gathered functional requirements into a cohesive digital environment. It commenced with the creation of low-fidelity wireframes (Appendix H) that mapped screens to the objectives and proposed features established in the Requirements Planning Phase. These wireframes served as a foundation for the system that allowed for the validation of user flow before moving into actual implementation.

The system follows a client-server, multi-tier architecture consisting of a Presentation Layer (Front-end), Business Logic Layer (Back-end), and Data Layer (Database). The front-end layer is developed using Tailwind CSS, React.js, and HTML5, providing a responsive design for user-centered access and navigation. Meanwhile, the back-end utilizes a hybrid environment where PHP handles core administrative functions such as grading logic, retention monitoring, and user management; meanwhile, the facial recognition attendance module is facilitated by Python-based OpenCV. Finally, the system utilizes a relation database schema hosted via MariaDB, comprising 14 tables, with relationships defined through strict foreign key constraints to maintain referential integrity.

Components interact through RESTful API calls; when the Python layer identifies an enrolled student, it transmits a timestamped log to the server. This update is then immediately reflected on the React.js-based dashboard, providing immediate feedback to the secretary and faculty members.

In addition, the following diagrams were constructed and are attached as appendices: the Context Diagram and Level 1 Data Flow Diagram (Appendix E), which illustrate the data flow between DentiSys and its external entities; the Use Case Diagram (Appendix F), which models system functionality using actors and use cases; and the Entity Relationship Diagram (ERD) (Appendix G), which demonstrates the relationships between core database entities.

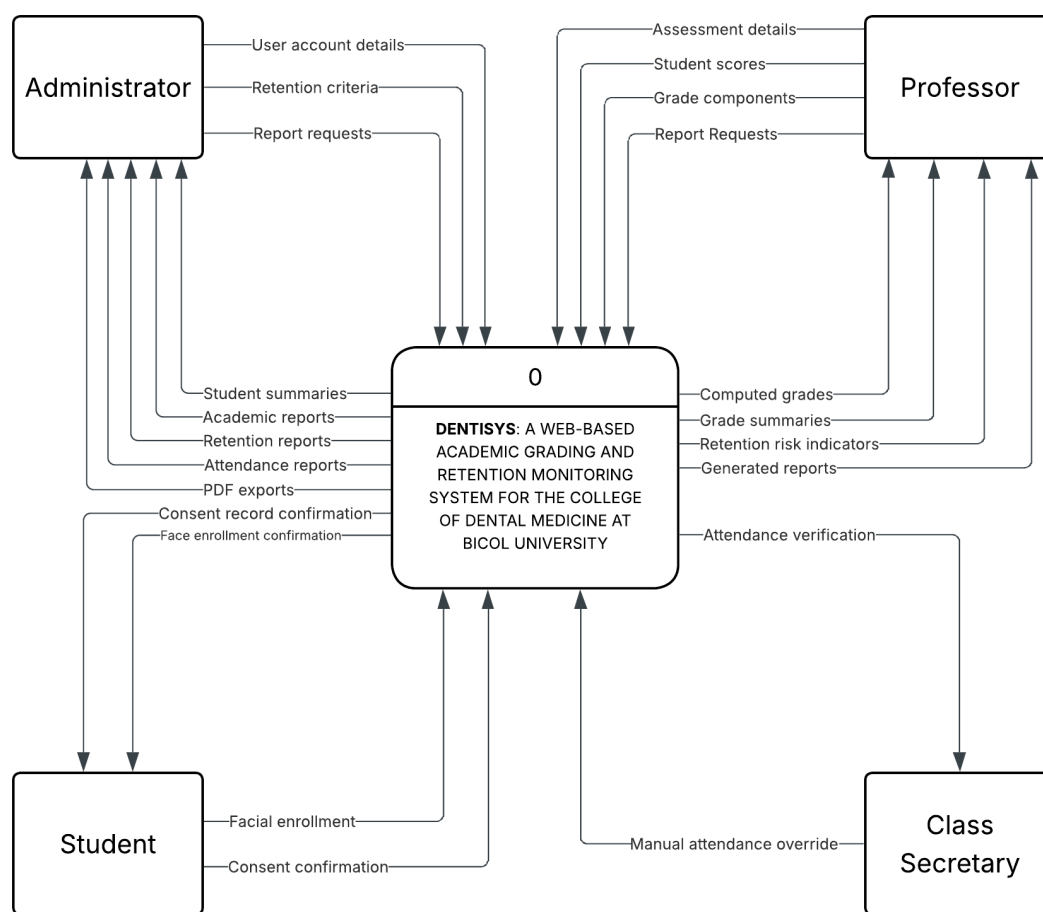


Figure 3. Data Flow Diagram (Context Diagram)

The Context Diagram, as shown in Figure 3, represents DentiSys as a single system process. It illustrates how it exchanges data with its external entities. These

external entities include the Administrator, Professor, Student, and Class Secretary. The Administrator provides user account details, retention criteria, and report requests, while receiving student summaries, academic reports, retention reports, attendance reports, and PDF exports. The Professor provides assessment details, student scores, grade components, and report requests while receiving computed grades, grade summaries, retention risk indicators, and generated reports. The Student provides facial enrollment data and consent confirmation, while the system returns confirmation for both the facial enrollment and the consent record. The Class Secretary provides manual attendance override if necessary, while the system returns the attendance verification result.

The Level 1 Data Flow Diagram presented in Appendix E further expands the Context Diagram by splitting DentiSys into its major internal processes. These processes include managing user access and system setup, managing academic records, processing grade computation, processing facial enrollment and attendance, monitoring retention status, and generating reports. Furthermore, it also illustrates the data stores used by the system, including user accounts, student records, course and section records, assessment records, grade records, facial data and consent records, attendance records, and retention criteria and status records. This further expansion of the Context Diagram provides a more detailed view of the data flow, showing how inputs from external entities are processed, stored, retrieved, and transformed into system outputs.

The use case diagram illustrates how three actors, the faculty, Class Secretary, and Admin/Dean, interact with the system, all starting with a login. The Faculty handles the most tasks such as entering grades, monitoring retention, managing attendance settings, customizing grading weights, and generating reports. The Class Secretary is

mainly responsible for enrolling student face data and manually marking attendance. The Admin/Dean oversees the system by viewing a system-wide summary and managing user accounts.

The entity relationship diagram (ERD) presented in Appendix F illustrates the relationships between the entities involved in the proposed system's grading, attendance, and retention monitoring modules. The Student entity acts as the central component that connects enrollment, assessment, attendance, and retention records. Additionally, the system includes entities for managing courses, class sections, user accounts, and facial recognition-based attendance, which is supported through the student image and facial template records.

3.2.3 Construction

The Construction Phase commences the development and coding of the DentiSys modules. This phase entails translating designs from the User Design phase into a working web-based application using previously established hardware and software resources.

The front-end construction began with HTML5 to structure the system and ultimately present content on the web. To bring this essential interface skeleton to life, Tailwind CSS and JavaScript were used. Tailwind CSS allowed the researchers to build the custom interface of DentiSys to accommodate rapid revisions and adjustments, while JavaScript ensured a resulting dynamic and interactive interface.

The developed user interface was then further bolstered by underlying structures: the back-end and database. To construct the database, the logical database design was

used as a blueprint to map out the relational schema within MariaDB. Each table was meticulously constructed with specific data types and constrained such that student records, attendance logs, and academic scores maintain entity, domain, and referential integrity. In order to make use of this database and implement core administrative functions, PHP was utilized. Its tasks involve session handling, managing user input, calculating grades, enforcing BUCDM's retention policies, exporting Excel files from the system, retrieving data from the database, and storing it to its appropriate locations. Furthermore, to handle the attendance module of the system, Python, utilizing OpenCV, was implemented. This dedicated service processed live camera feeds and student identification using the LBPH algorithm without slowing down the web application or requiring high system specifications.

To ensure the system's reliability and alignment with BUCDM's standards, extensive testing occurred. Prior to the final client evaluation, the developers conducted inspections to ensure that every module conformed strictly to defined requirements. DentiSys was subjected to various tests to address functionality, scalability, performance, correctness, and security.

3.2.4 Cutover

The cutover phase is the final phase of development that facilitates the deployment of the DentiSys system. In this phase, the preparations done during the construction phase were tested and evaluated for the final time to ensure system readiness for deployment. The final phase of testing included checking the client-server connections, module integration checks, testing of module functionality, output validation

checks, and overall usability testing. Once the final rounds of testing were completed, the completed system was shown to faculty and students for user acceptance testing.

During user acceptance testing, the system was evaluated by the Dean, Assistant Deans, and Block secretaries to see if they meet the necessary requirements illustrated in the Requirements Phase of development. The system was assessed using a five-point Likert scale that is based on certain ISO/IEC 25010 metrics; specifically in terms of functional suitability, reliability, performance efficiency, security, and usability. Furthermore, this phase is also where the users of the DentiSys were briefed with how to utilize the system and its different modules.

Upon completion of user testing, the researchers implemented necessary changes and improvements based on the feedback and results of the evaluation. Testing and iteration was done until the system fully complied with user requirements. Once the system was deemed functionally stable, and met user demands, the system was confirmed operational and ready for deployment.

3.3 Evaluation Procedure

To evaluate the system, the researchers will use the ISO/IEC 25010 software quality model as the basis for the survey questionnaire. The evaluation will focus on functional suitability, portability, reliability, maintainability, compatibility, performance efficiency, security, and usability, since these areas are directly related to the system's main functions, such as grade computation, attendance monitoring, retention tracking, and report generation (International Organization for Standardization, 2023).

The respondents will be selected from the College of Dental Medicine of Bicol University because they are the intended users of the system. Faculty members will evaluate the parts of the system related to grade encoding, attendance monitoring, and report generation, while student respondents will assess the features available to them. Purposive sampling will be used to ensure that the participants have direct experience with the academic processes covered by the system (Etikan et al., 2016).

Table 4. Likert Scale

Scale	Range	Meaning	Description
5	4.21-5.00	Strongly Agree	The system has functional completeness of tasks, is effective and reliable.
4	3.41-4.20	Agree	The system has somehow functional completeness of tasks, is somehow effective and reliable.
3	2.61-3.40	Neutral	The system has slightly functional completeness of tasks, is slightly effective and reliable.
2	1.81-2.60	Disagree	The system has slightly no functional completeness of tasks, is slightly not effective and reliable.
1	1.00-1.80	Strongly Disagree	The system has no functional completeness of tasks, is not effective and reliable.

Prior to answering the questionnaire, respondents will be given access to the system and asked to perform basic tasks based on their assigned roles. After interacting with the system, they will rate each survey item on a five-point Likert scale. The items will cover areas such as grade computation accuracy, system response time, reliability, data security, and ease of use (Jebb et al., 2021).

The interpretation of the weighted mean was based on the Likert scale shown in Table 4. The scale was used to classify the computed means according to respondents' levels of agreement, which ranged from Strongly Disagree to Strongly Agree.

The weighted mean was computed using the formula:

$$M = \frac{\sum fx}{N}$$

where:

- f = frequency of responses
- x = assigned Likert scale value
- N = total number of responses

The results of the computation were used to assess the system's acceptability against the selected ISO/IEC 25010 criteria.

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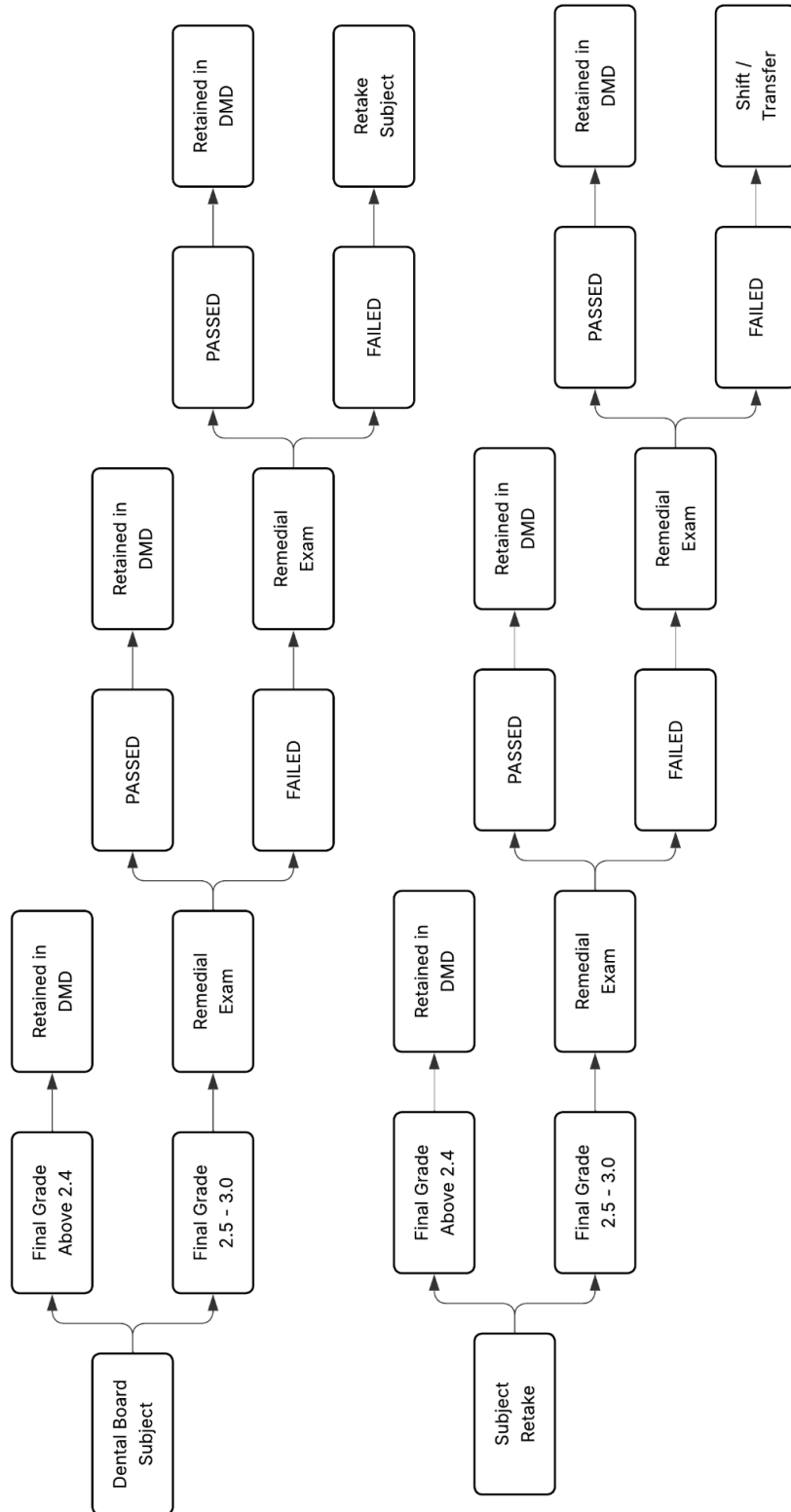
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APPENDICES

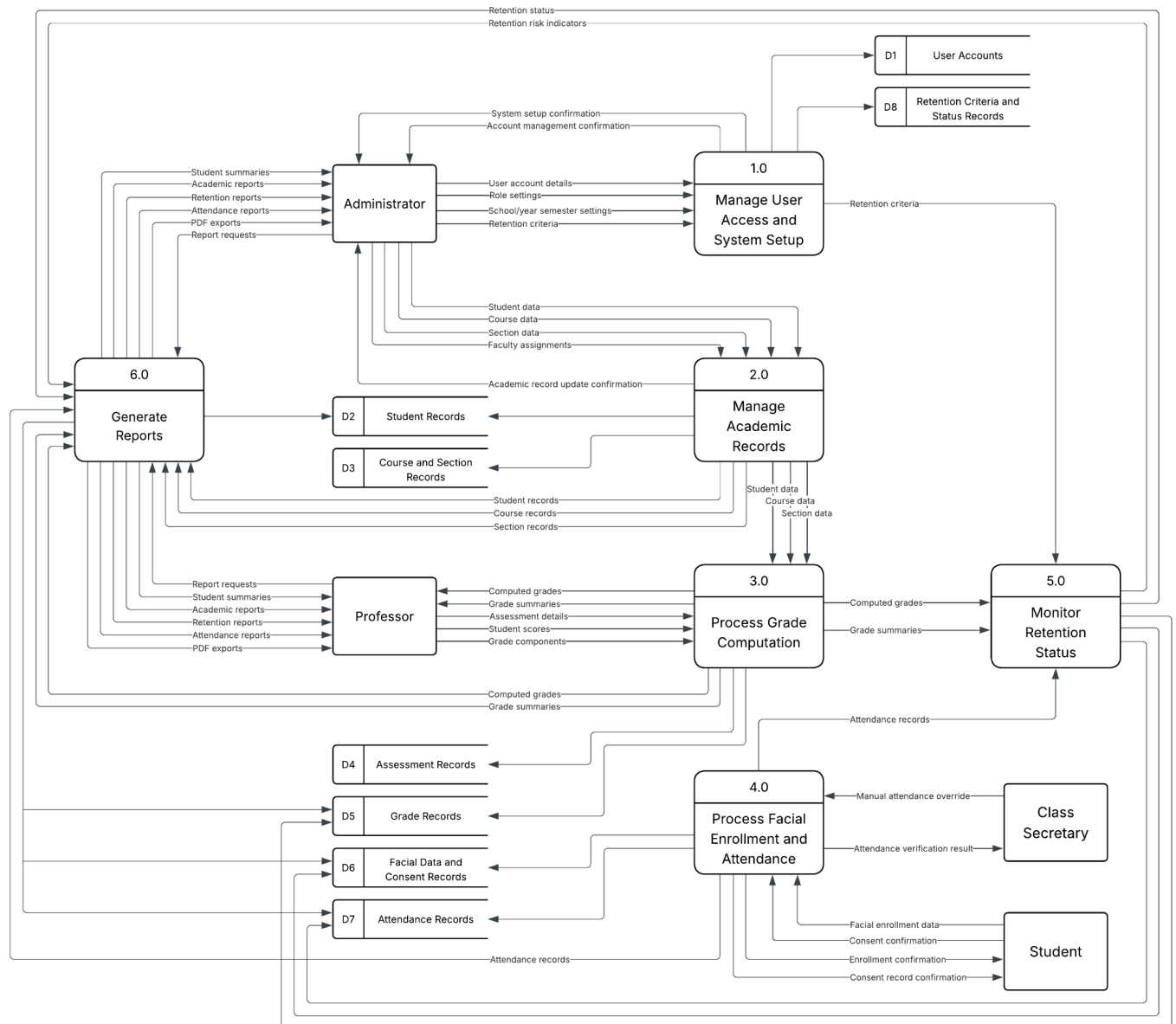
Appendix A: Letter of Request

Appendix B: Appointments

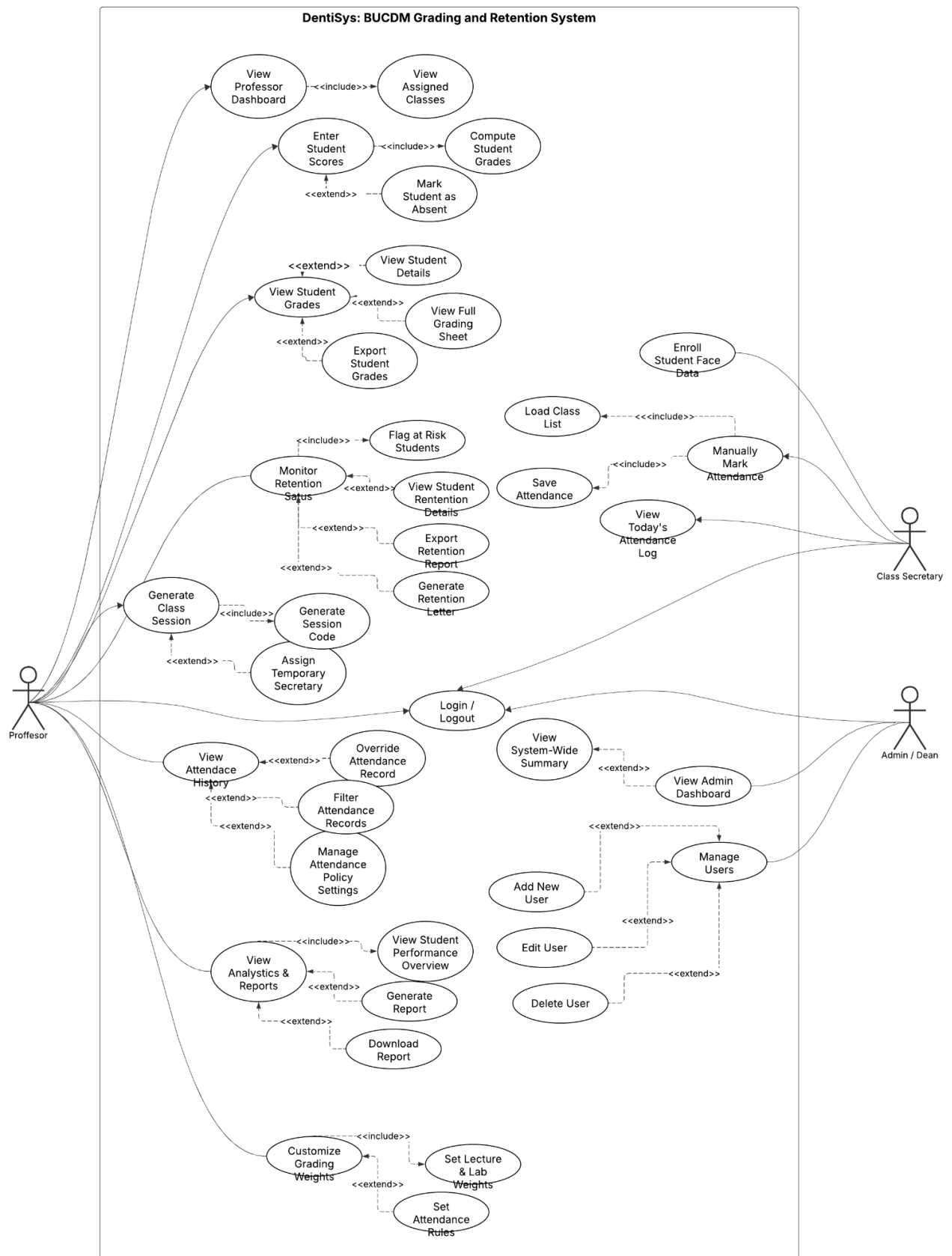
Appendix C: BUCDM's Retention Policy Flowchart

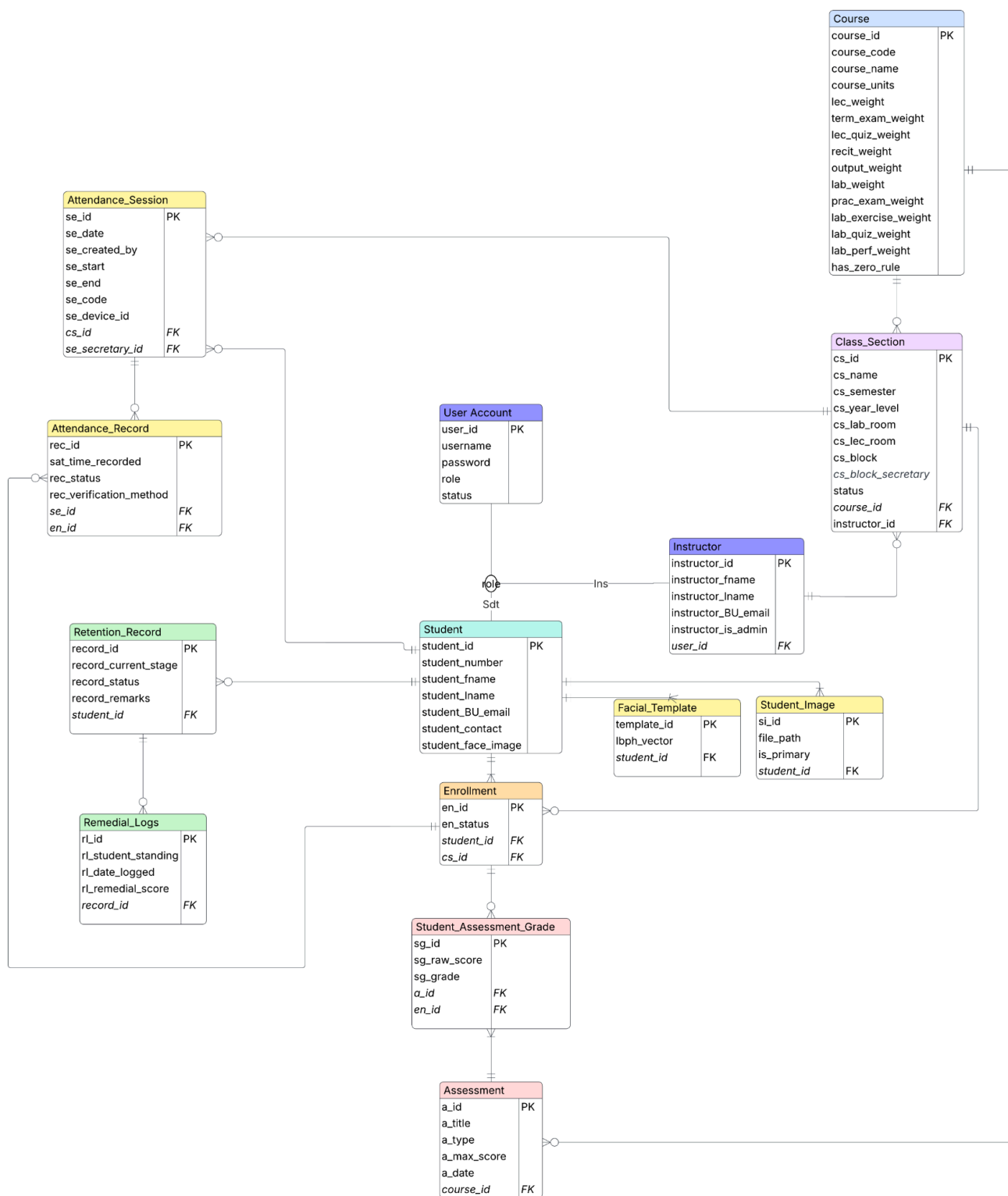


Appendix D: Data Flow Diagram (Level 1)



Appendix E: Use Case Diagram of the Proposed System





Appendix G: Low-fidelity Wireframes of the Proposed System

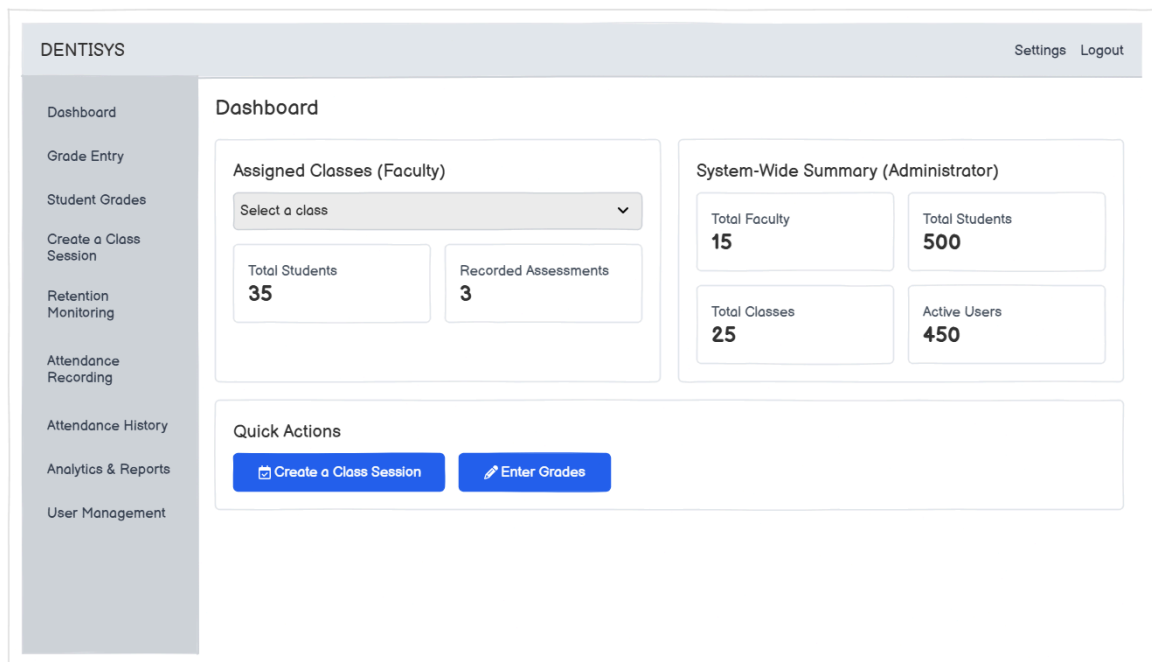


Figure H1. Faculty-facing Dashboard

Figure H1 shows a faculty-facing dashboard that serves as the main landing page for faculty upon logging into the DentiSys system. It provides a quick overview of class information, attendance records, and retention alerts. In addition, it displays quick action buttons for easier access to day-to-day operations.

DENTISYS
Settings Logout

Dashboard
Grade Entry
Student Grades
Create a Class Session
Retention Monitoring
Attendance Recording
Attendance History
Analytics & Reports
User Management

Grade Entry

Course:

Block:

Date:

Component (Lab/Lec):

General Anatomy

2A

5 / 2 / 2026

Lab

Activity:

Quiz

Total Points:

Description: (Optional)

10

Quiz on the Skeletal System

Student ID	Student Name	Score	Grade
S001	Mark Grayson		
S002	Adam Wilkins		
S003	Angstrong Levy	Absent	0

Save Scores

Figure H2. Grade Entry Interface

As shown in Figure H2, the grade entry interface allows faculty members to input, update, and manage student grades per assessment. It shows a tabular layout to resemble their prior Excel-based system, sans the manual cell manipulation. It shall automatically calculate assessment grades based on defined grading weights and attendance records.

DENTISYS
Settings Logout

Dashboard
Grade Entry
Student Grades
Create a Class Session
Retention Monitoring
Attendance Recording
Attendance History
Analytics & Reports
User Management

Student Grades

Select A.Y.: Select an option
Select Semester: Select an option
Select Term: Select an option

Select Class: Select an option

View Grades

Student Name	Lecture Grade (60%)	Laboratory Grade (40%)	Final Grade	Remarks	Actions
Mark Grayson	85.00	90.00	2.2	PASSED	View Details
Adam Wilkins	70.00	75.00	72.00	SUBJECT FOR REMEDIAL	View Details
Angstrong Levy	99.00	90.00	1.4	PASSED	View Details

View Full Grading Sheet

Export final consolidated grades

Export to Excel

Export to PDF

Figure H3. Student Grades Interface

The student grade interface in Figure H3 exhibits a consolidated view of student grades. It enables faculty to review computed grades and export both consolidated and comprehensive grading sheets in multiple formats.

DENTISYS

Back to DashboardLogout

Manage Courses
Manage Classes
Grading Weights
Secretary
Assignment

Customize Course Grading Weights

Select Course: Select course ▼

Lecture: %

Term exam: %

Quiz: %

Recitation: %

Output: %

Laboratory: %

Practical exam: %

Lab exercises: %

Quiz: %

Lab performance: %

Attendance Rules

Toggle rule that absence resolves to zero (0) quiz grade.

☒

Save Settings

Figure H4. Course Grading Weight Customization

Figure H4 shows the interface specifically designed for customizing grading weights per course.

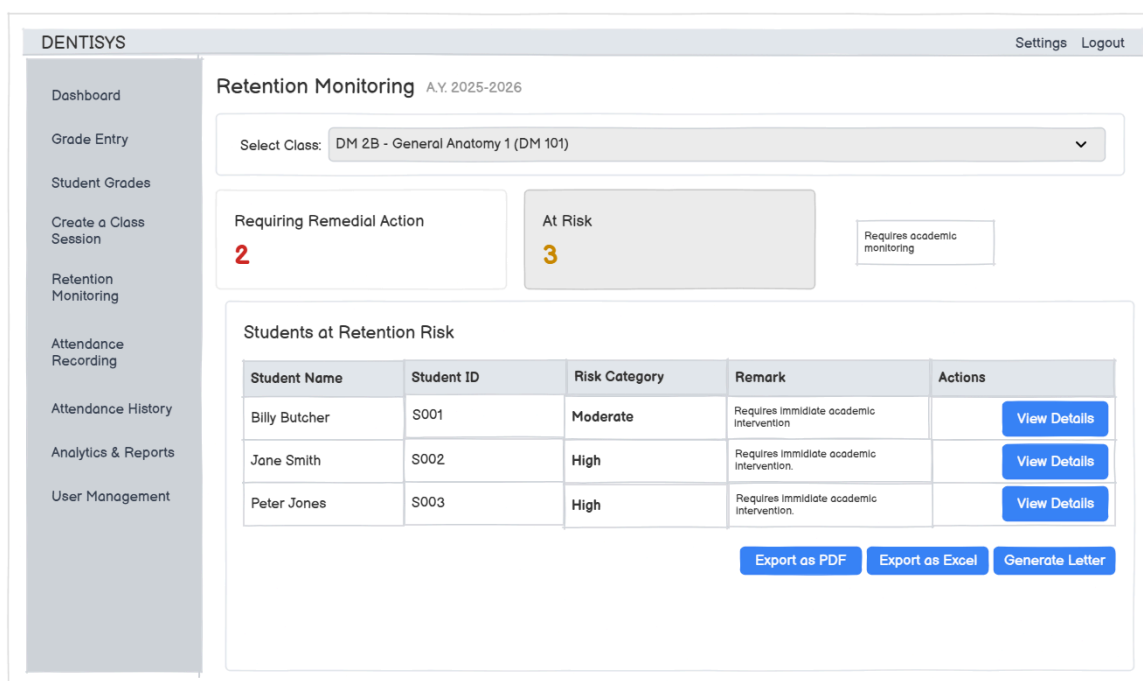


Figure H5. Monitoring Retention Risk

Monitoring retention risk is illustrated by Figure H5; the interface provides faculty members with insights into which students may be at risk of failing retention requirements. The wireframe presents students that are at moderate and high risk of requiring remedial action or failing retention requirements, which give way for proactive decision-making and early student support.

DENTISYS
Settings Logout

Dashboard
Grade Entry
Student Grades
Create a Class Session
Retention Monitoring
Attendance Recording
Attendance History
Analytics & Reports
User Management

Retention Monitoring A.Y. 2025-2026

Select Class: DM 2B - General Anatomy 1 (DM 101) ▼

Requiring Remedial Action
2

At Risk
3

Students Requiring Remedial Action

Student Name	Student ID	Status	Remark	Actions
Billy Butcher	S009	PASSED	Passed First Remedial Exam	Update View Details
Hughie Campbell	S008	PENDING	Subject for Second Remedial Exam	Update View Details

Export as PDF
Export as Excel
Generate Letter

Figure H6. Monitoring Remedial Action

As seen in Figure H6, the remedial action monitoring interface is designed to track students who require immediate intervention or are currently undergoing it. It enables faculty to record and monitor student conformance to retention requirements.

The screenshot displays the DENTISYS web application interface. At the top, a header bar contains the 'DENTISYS' logo on the left and 'Settings' and 'Logout' links on the right. A vertical sidebar on the left lists navigation options: Dashboard, Grade Entry, Student Grades, Create a Class Session, Retention Monitoring, Attendance Recording, Attendance History, Analytics & Reports, and User Management. The main content area is titled 'Dashboard' and features a 'Class Sessions' section. This section includes two buttons: 'Create a Class Session' (highlighted in blue) and 'View Scheduled Sessions'. Below these buttons is a form titled 'Create a class session for'. The form contains two dropdown menus labeled 'Class:' and 'Room:', each with a 'Select Class' and 'Select Classroom' option respectively. Below these are two time input fields labeled 'FROM' and 'UNTIL', both showing '5:45 PM'. A checkbox labeled 'Repeat' is positioned below the time fields. At the bottom of the form are two buttons: an orange button labeled 'Assign a Temporary Class Monitor' and a blue button labeled '✓ Create Class Session'.

Figure H7. Creating a Class Session

The interface for creating a class session can be seen in Figure H7. This interface enables instructors to establish attendance sessions for scheduled classes. Faculty members can configure details such as course section, classroom, date, time, and attendance settings before initiating the session.

DENTISYS
Manually Mark Attendance
Logout

Attendance Recording

Today's Attendance Log

Student Name	Student ID	Time	Status	Method
Mark Grayson	S001	08:00 AM	Present	Facial Rec.
Adam Wilkins	S002	08:05 AM	Present	Facial Rec. (Override)
Amber Bennet	S004	08:10 AM	Present	Manual
Angstrom Levy	S003		Absent	
Unknown				Facial Rec.

Facial Recognition Attendance



Camera Feed / Detected Unknown Face

Manual Override / Student Details

Student ID:

Student Name:

Attendance Status:

Select Status
▼

Apply Override

Figure H8. Student-side Attendance Recording

Figure H8 displays the student-side interface of attendance recording with facial recognition. It allows the assigned class monitor/s to review attendance records for any discrepancies and amend them accordingly. It facilitates a Human-in-the-Loop (HITL) approach that allows any system inaccuracies to be rectified through manual overrides and facial re-enrolment.

DENTISYS
Settings Logout

Dashboard
Grade Entry
Student Grades
Create a Class Session
Retention Monitoring
Attendance Recording
Attendance History
Analytics & Reports
User Management

Attendance History

Filter Attendance Records

Student Name
Date From
Date To

Course
Section
Status

Attendance Records

Student Name	Date	Time In	Time Out	Status	Taken through	Actions
Mark Grayson	2025-10-26	08:00 AM	05:00 PM	Present	Facial Recognition	Override
Angstrong Levy	2025-10-26	-	-	Absent	-	Override
Adam Wilkins	2025-10-25	08:30 AM	05:00 PM	Present	Manual override (click for image)	Override

Policy Settings

Zero Quiz Score for Absences: Enabled

Figure H9. Class Attendance History

The class attendance history wireframe shown in Figure H9 displays a record of previous attendance sessions. Through this interface, faculty members can review attendance logs, enforce Bicol University's minimum attendance policy, and override attendance statuses.